

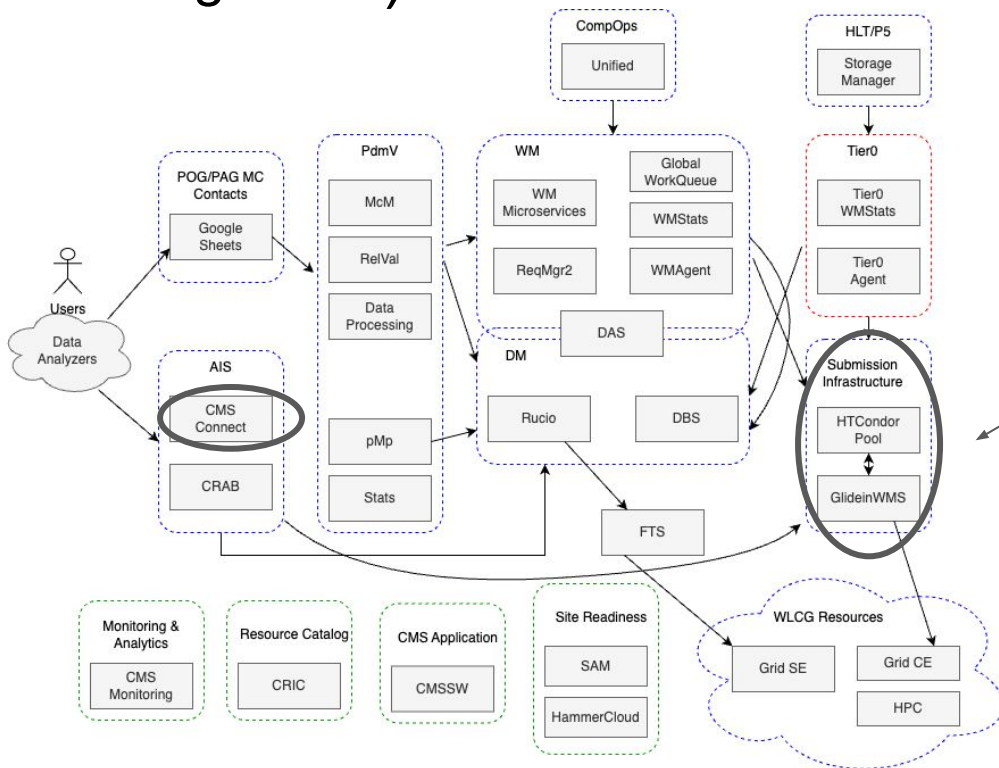
GPU HEP-Workloads running on grid (CMS)

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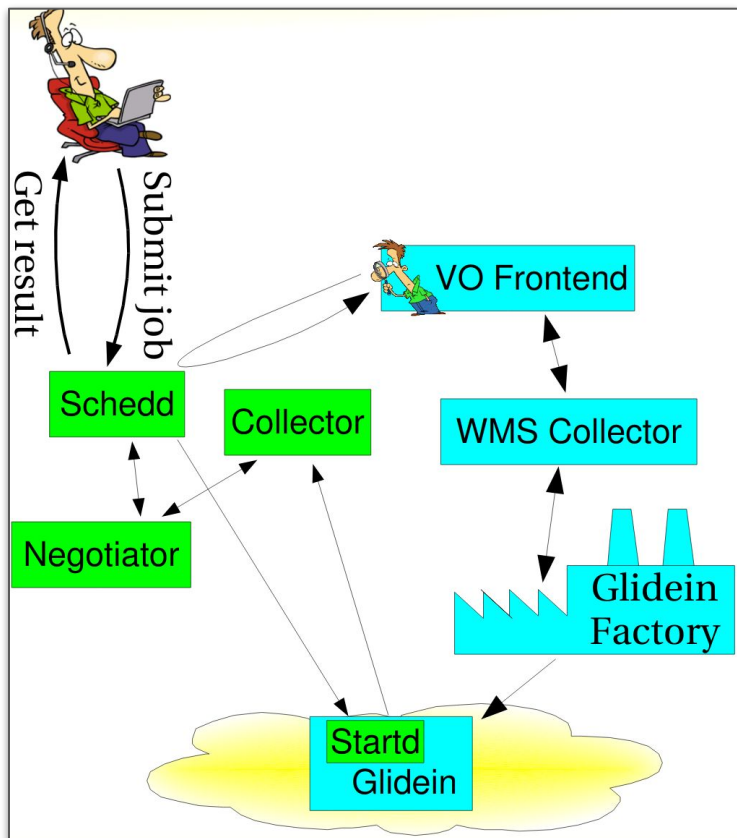


Overall CMS DMWM (Data Management & Workflow Management)



HEPScore tests run through **CMSScore**: light service that enables users to **submit** to all resources available in the **CMS Global Pool**

CMSScore and GlideinWMS allow uniform access to heterogeneous resources, including GPUs.



Two matchmaking stages in Submission Infrastructure: resource allocation (GlideinWMS) and job to slot matchmaking (HTCondor)

1. GlideinWMS and pilot jobs

- Submit resource requests on sites CEs in order to join them into the Global HTCondor Pool

2. HTCondor matchmaking

- Slot joins the Global Pool, then resources are negotiated and assigned to payload jobs

Heterogeneous resources just follow the same general logic

See backup slide for details about submission

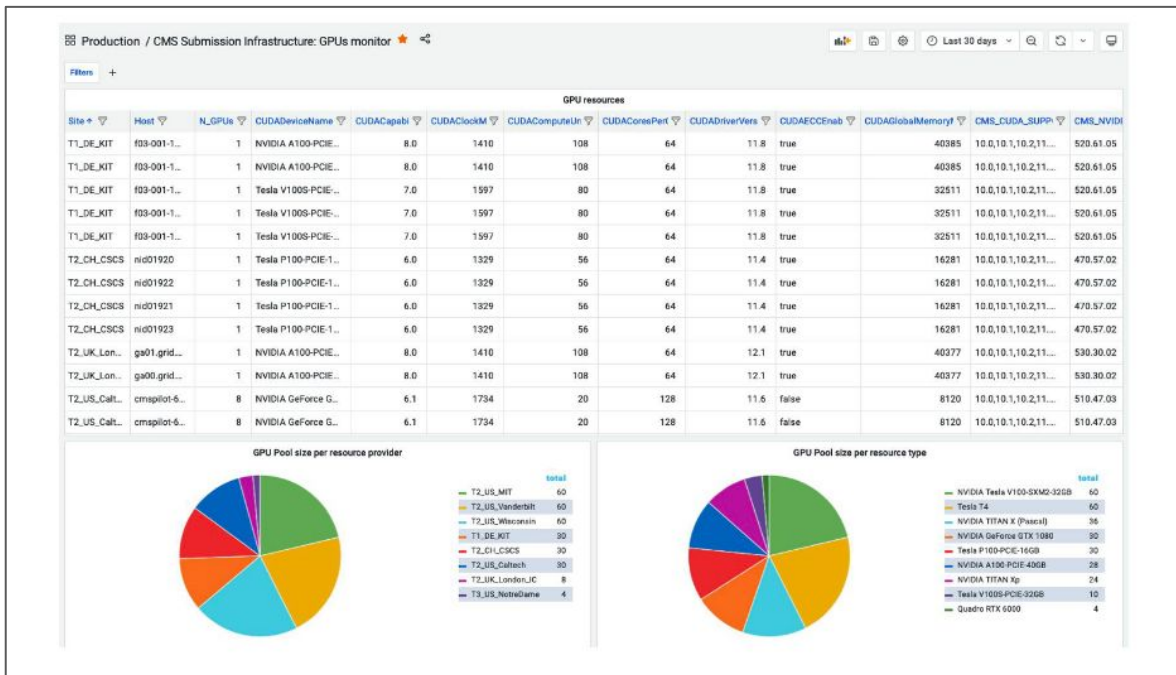
Integration of GPU resources in CMS

- **GPU resources** in the CMS global pool since ~**2022** (see CHEP23 [ref](#))

The Global Pool FE submits **scout pilots** to GPU-like CEs present in our pilot factories

HTCondor GPU discovery tool allows to **gather information** about the devices

The SI team has built an internal [catalogue of GPU slots](#) to inform users about “what is available and where”



GPU device information in the pilot classads

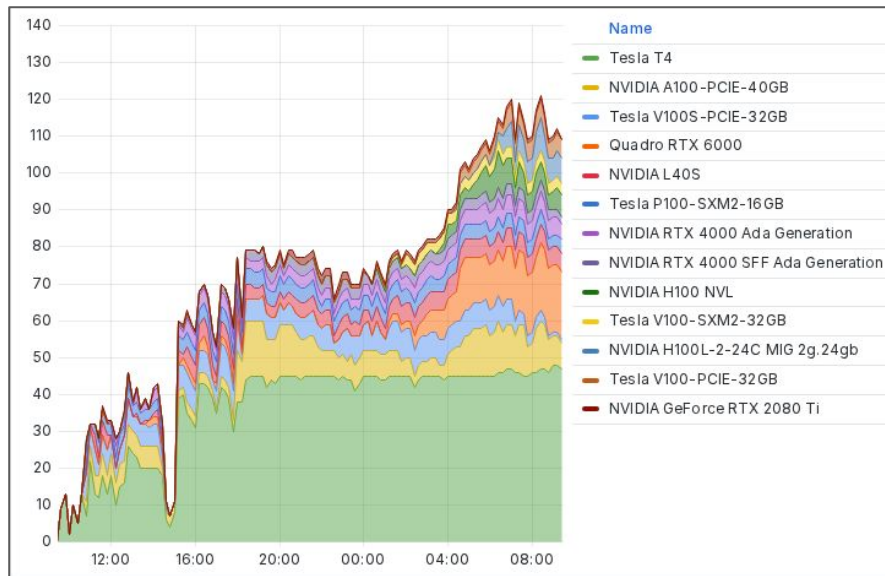
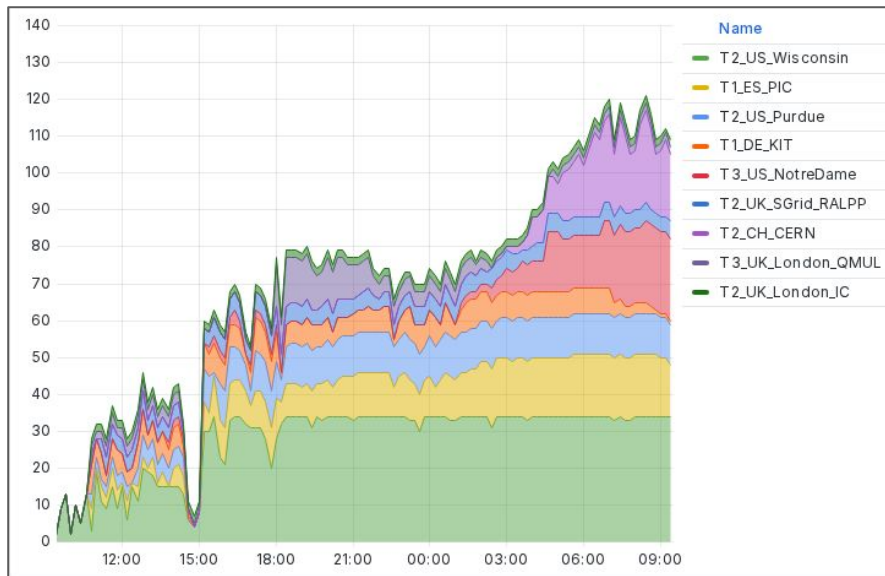
- Our Global Pool startds are running latest stable release in HTCondor v24
- Pilots execute the [condor gpu discovery](#) tool to gather device information, for example:

```
GPUs = 1
GPUs_Capability = 7.0
GPUs_ClockMhz = 1597.0
GPUs_ComputeUnits = 80
GPUs_CoresPerCU = 64
GPUs_DeviceName = "Tesla V100S-PCIE-32GB"
GPUs_DevicePciBusId = "0000:01:00.0"
GPUs_DeviceUuid = "17fc2977-eee9-80df-4da2-e5bdf8571f4"
GPUs_DriverVersion = 13.0
GPUs_ECCEnabled = true
GPUs_GlobalMemoryMb = 32494
GPUs_GPU_17fc2977 = [ Id = "GPU-17fc2977"; ClockMhz = 1597.0; Capability = 7.0; CoresPerCU = 64; DeviceName =
"Tesla V100S-PCIE-32GB"; DeviceUuid = "17fc2977-eee9-80df-4da2-e5bdf8571f4"; ECCEnabled = true; ComputeUnits
= 80; DriverVersion = 13.0; DevicePciBusId = "0000:01:00.0"; GlobalMemoryMb = 32494; MaxSupportedVersion =
13000 ]
GPUs_MaxSupportedVersion = 13000
```

This metadata is exposed
in the ClassAds, allowing
GPU-aware matchmaking.

CMS GPU Availability Snapshot (O&C Week 2025)

- During CMS O&C week we run a **mini test** in the Global Pool to get an overview of the current availability of GPUs in our grid
 - Max ~120 GPU slots in about 24h



CMS GPU Availability Snapshot (O&C Week 2025)

- An quick overview table of what we have in the pool

Site	Distinct Hosts	GPU Models	GPUs per Host (range)	Total GPUs
T1_DE_KIT	1	Tesla V100S-PCIE-32GB	8	8
T3_UK_London_QMUL	7	NVIDIA A100-PCIE-40GB	2	14
T2_UK_London_IC	2	NVIDIA A100-PCIE-40GB	1-2	3
T2_UK_SGrid_RALPP	2	P100-SXM2-16GB, A100-PCIE-40GB	1-4	5
T1_ES_PIC	4	RTX 4000 Ada, RTX 4000 SFF Ada, L40S	2-4	11
T2_US_Purdue	11	Tesla T4	1	11
T2_US_Wisconsin	17	Tesla T4	2	34
T3_US_NotreDame	1	Quadro RTX 6000	1	1

More serious GPU benchmarking

- Multiple GPU-like workloads available in the HEPIX Benchmarking suite (from multiple-VO)
 - Some CMS workloads already included (Robin's [talk](#) Oct 1.)
- CMS-Flowsim-GPU was chosen for the new benchmarking test
 - Most stable one and the best understood, good candidate for a workload that may actually run on the grid

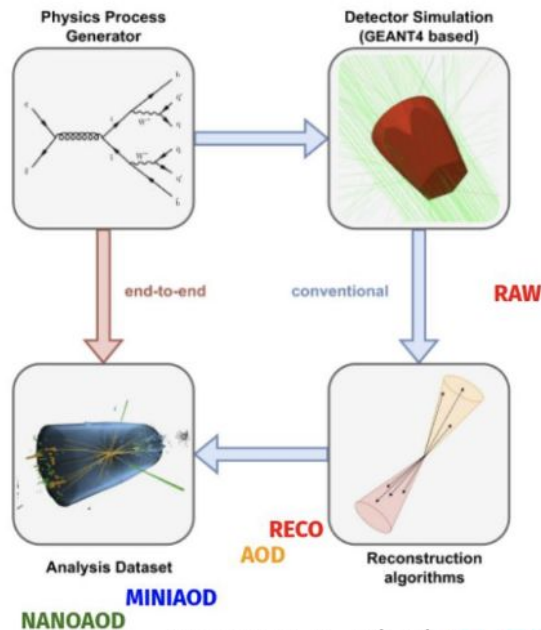
Workload Overview

GPU Workloads for HEPsScore

Workload	Type
CMS-HLT-GPU	Reconstruction
Pepper-GPU	Simulation
CMS-Flowsim-GPU*	Inference
CMS-MLPF-GPU*	Training
Madgraph-GPU	Simulation
Alice-O2-GPU	Reconstruction
Synthetic-GPU	Synthetic+ML(?)
ET-RIFT	--External--
CMS-HLT-MA	Reconstruction
madgraph4gpu-2022	Simulation
CMS-HGCAL**	Inference

* ML-based WLS

** Candidate, to be evaluated



“End-to-end simulation of particle physics events with flow matching and generator oversampling”

- [Repo](#)
- [Paper](#)

Benchmarking test for GPUs in CMS SI

- Run test jobs on GPUs using this benchmark as payload in the Global Pool

```

////////////////////////////////////
//                               //
//          +++++ Attention **** //
//-----
// CMS-Flowsim is currently only available for NVIDIA GPUs. //
// AMD support will be added soon! //
////////////////////////////////////

```

```

"host": {
  "hostname": "gpu06.pic.es",

```

```

},
"GPU": {
  "nvidia0": {
    "name": "NVIDIA RTX 4000 Ada Generation",
    "memory_total": "20475 MiB",
    "memory_used": "4 MiB",
    "clock_graphics": "210 MHz",
    "clock_sm": "210 MHz",
    "pci_bus": "00000000:21:00.0",
    "power_avg": "11.30 W"
  }
}

```



```

Total throughput: 11365.88 objects/second
Total time: 791.84s
Total objects: 9000000

```

```

Resource Usage Summary:
CPU Average: 63.3%
CPU Peak: 71.0%
RAM Average: 24.1%
RAM Peak: 24.9%
GPU Utilization Average: 99.0%
GPU Utilization Peak: 100.0%
GPU Memory Average: 91.8%
GPU Memory Peak: 98.7%

```

Integrating HEPSCORE test on the grid

Few minor issues faced while running hepscore on the Grid

- User namespace not available while running the test (?)

```

KIT failed with this error in HEPscore GPU.log:
INFO: Cleanup error: while unmounting /var/lib/apptainer/mnt/session/final directory: no such file or directory, while unmounting /var/lib/apptainer/mnt/session/rootfs directory: no such file or directory
FATAL: container creation failed: multiple devpts instances unsupported and /dev/pts configured
    
```

- GPU does not have enough memory to run the test

```

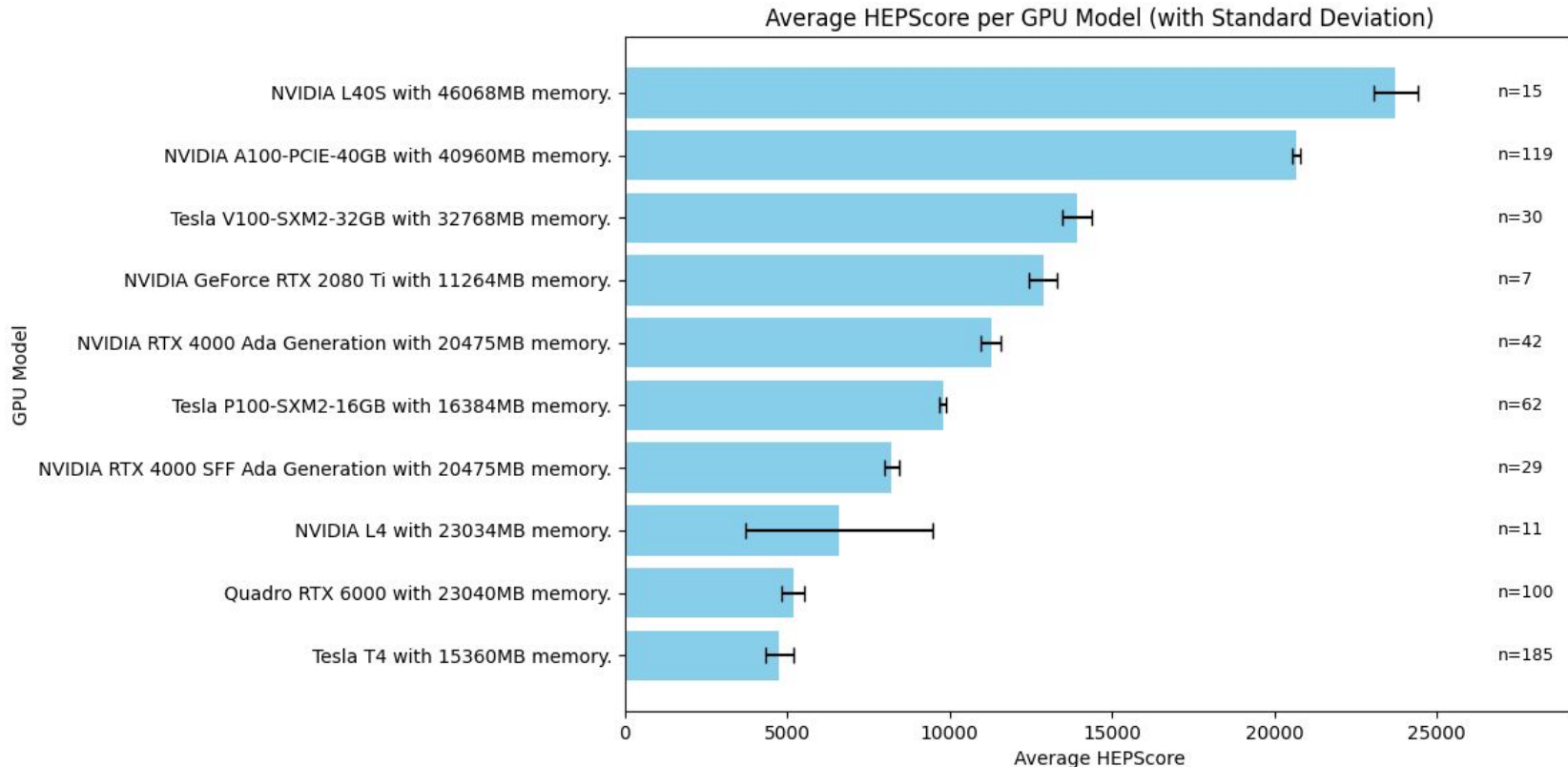
Error during benchmark: CUDA out of memory. Tried to allocate 1.05 GiB. GPU 0 has a total capacity of 14.56 GiB of which 683.81 MiB is free. Including non-PyTorch memory, this process has 13.89 GiB memory in use. Of the allocated memory 13.73 GiB is allocated by PyTorch, and 36.48 MiB is reserved by PyTorch but unallocated. If reserved but unallocated memory is large try setting max_split_size_mb to avoid fragmentation. See documentation for Memory Management and PYTORCH_CUDA_ALLOC_CONF
Run on Tesla T4 with 14913 Mb of ram
    
```

- Environment pollution causes openssl to user wrong library

```

File "/srv/workdir/env_bmk/lib64/python3.9/site-packages/OpenSSL/__init__.py", line 8, in <module>
    from OpenSSL import SSL, crypto
File "/srv/workdir/env_bmk/lib64/python3.9/site-packages/OpenSSL/SSL.py", line 42, in <module>
    from OpenSSL.crypto import (
File "/srv/workdir/env_bmk/lib64/python3.9/site-packages/OpenSSL/crypto.py", line 2395, in <module>
    utils.deprecated()
TypeError: deprecated() got an unexpected keyword argument 'name'
tar: Removing leading '/' from member names
tar: /srv/workdir/suite.results: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
    
```

Preliminary HEPScore-GPU Benchmark Results



Other GPUs available

☐	1001027	Request for GPU resource verification - HEPSCORE benchmarking tests	Vanderbilt	CMS Glide-In Factory	cms	Marco Mascher...	USCMS	assigned	2025-10-31 17:41	2025-10-31 17:41
☐	1001026	Request for GPU resource verification - HEPSCORE benchmarking tests	Nebraska	CMS Glide-In Factory	cms	Marco Mascher...	USCMS	assigned	2025-10-31 17:39	2025-10-31 17:39
☐	1001025	Request for GPU resource verification - HEPSCORE benchmarking tests	MIT_CMS	CMS Glide-In Factory	cms	Marco Mascher...	USCMS	assigned	2025-10-31 17:38	2025-10-31 17:38
☐	1001024	Request for GPU resource verification - HEPSCORE benchmarking tests	UFlorida-HPC	CMS Glide-In Factory	cms	Marco Mascher...	USCMS	assigned	2025-10-31 17:37	2025-10-31 17:37
☐	1001023	Request for GPU resource verification - HEPSCORE benchmarking tests	CIT_CMS_T2	CMS Glide-In Factory	cms	Marco Mascher...	USCMS	assigned	2025-10-31 17:36	2025-10-31 17:36

- GPU resources are opportunistic. Many other sites provided GPUs at some point
 - Opened ticket to investigate why we can't access them anymore
 - Seems like GPUs got re-routed due to low utilization. E.g.:
 - GPU resources at Nebraska have been removed from the T2 and placed in analysis facility
 - MIT does not provide anymore
 - Vanderbilt updated cluster, GPUs moved to a different partition. Trying replacement
- Investigating issues at other sites
 - Bari, Caltech, Syracuse
 - Florida fixed this morning
- Non Grid resources (HPC) that can't run user jobs

Challenges using GPUs in SI

Remaining challenges include

- **GPU benchmarking** and accounting **required for realistic usage at scale in the WLCG context**:
 - **Benchmarking** GPUs for CMS workloads is a requirement for pledge process and resource acquisition
 - **Predictable workflow runtimes**, a key parameter for an efficient matchmaking of jobs to slots
 - hard because of high heterogeneity among GPUs
- **Efficient execution** of CMS multi-steps jobs on **CPU+GPU resources**
 - Not only in WLCG, but also at current and future HPCs
- Support for non-NVIDIA GPUs still to be exercised

Conclusions

- **Dealing with resource heterogeneity** in CMS computing is a **key element** moving forward
- **No standard GPU slot definition exists**, workload scheduling requires careful description of slot properties and workload requirements for effective matchmaking
- With regards to **GPUs**, still mostly **opportunistic** but a sizable number and variety **already available** in the CMS
- Policies for resource acquisition and matchmaking are not trivial, some decisions are needed
- Challenges ahead
 - Efficient execution of CMS multi-steps jobs on CPU/GPU heterogeneous resources
 - Benchmarking (and accounting)
 - Non-NVidia GPUs

Backup Slides

Access CMS Global Pool to run on GPU nodes

Can easily access CMS Global pool and run tests on GPU resources from CMS Connect (CMS users only) or by requesting an idtoken to the Submission Infrastructure group (e.g.: remote **condor_submit** from lxplus)

Universe = vanilla

Executable = **run.sh**

```
Log      = log/run.$(Cluster)
Output   = out/run.$(SITE).$(Cluster).$(Process)
Error    = err/run.$(SITE).$(Cluster).$(Process)
```

Job log

```
should_transfer_files = YES
when_to_transfer_output = ON_EXIT
transfer_input_files = run_HEPscore.sh,hepscore-gpu-flowsim.yaml
transfer_output_files = output.tgz
transfer_output_remaps = "output.tgz = results/output.$(SITE).$(Cluster).$(Process).tgz"
```

Job input/output

```
+DESIRED_Sites = "$$(SITE)"
request_cpus = 1
request_gpus = 1
+RequiresGpu = 1
request_memory = 2000
+MaxWallTimeMins = 180
+REQUIRED_OS = "rhel9"
```

Job specification

Queue 100 SITE from sitelist.txt

sitelist.txt

```
T1_DE_KIT
T1_ES_PIC
T2_CH_CERN
T2_IT_Bari
T2_UK_London_IC
T2_UK_SGrid_RALPP
T2_US_Caltech
T2_US_Florida
T2_US_Purdue
T2_US_Wisconsin
T3_UK_London_QMUL
T2_US_Vanderbilt
T3_US_NotreDame
T3_US_OSG
```